

Notes: Kisgen (JF, 2006)

Credit Ratings and Capital Structure

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1 Big picture

- **Question.** Do credit ratings directly affect firms' capital structure decisions, beyond the usual tradeoff and pecking-order determinants?
- **Core idea.** Credit ratings are discrete categories. Moving from one rating to another can trigger discrete costs or benefits through regulation, investor mandates, capital requirements, contracting clauses, and market perceptions.
- **CR-CS hypothesis.** Firms close to an upgrade or downgrade issue less debt relative to equity than otherwise similar firms not close to a rating change.
- **Main empirical measures.** The paper defines rating-change proximity in two ways: plus/minus ratings within broad ratings (POM test), and a credit-score ranking within micro ratings (Credit Score test).
- **Main dependent variable.** $\text{NetDIss} = (\Delta D - \Delta E)/A$, the net debt issuance minus net equity issuance, scaled by beginning-of-year total assets.
- **Sample.** Compustat firms with S&P long-term domestic issuer credit ratings over 1986–2001. The baseline sample contains 12,336 firm-years.
- **Main result.** Firms near a rating change issue roughly 1.0% less net debt relative to equity, as a fraction of total assets, than firms not near a rating change.
- **Why large offerings are treated separately.** Very large offerings can move almost any firm across rating boundaries. The main tests therefore exclude debt offerings greater than 10% of assets.
- **Robustness.** The results survive controls for leverage, profitability, size, broad ratings, explanatory variables from traditional capital-structure models, and nested tests of pecking-order and tradeoff theories.
- **Economic takeaway.** Credit ratings are not just labels attached after capital-structure choices. Managers appear to adjust financing decisions to avoid downgrades or to preserve the chance of upgrades.

2 Data and measurement

2.1 Credit ratings and sample construction

- The credit rating is the S&P Long-Term Domestic Issuer Credit Rating from Compustat.
- The sample begins in 1986 because 1985 is the first year in which this rating is available in Compustat.
- The sample ends in 2001 and includes firms with the variables needed for the financing regressions.
- Financial firms and utilities are included in the baseline because rating concerns can matter for them too; robustness tests exclude them.

Interpretation: The focus is on firms that already access rated debt markets. The results therefore speak directly to rated public issuers, not to unrated firms.

2.2 Capital-structure variables

- D_{it} is book long-term debt plus short-term debt.
- ΔD_{it} is long-term debt issuance minus long-term debt reduction plus changes in current debt.
- ΔLTD_{it} is long-term debt issuance minus long-term debt reduction.
- E_{it} is book shareholder equity.
- ΔE_{it} is sale of common and preferred stock minus purchases of common and preferred stock.
- A_{it} is beginning-of-year total assets.

The main financing outcome is

$$\text{NetDIss}_{it} = \frac{\Delta D_{it} - \Delta E_{it}}{A_{it}}. \quad (1)$$

Interpretation: A lower NetDIss means the firm finances more conservatively with respect to debt: less debt issuance, more equity issuance, or both.

2.3 Control variables

The baseline controls are collected in K_{it} :

- leverage: $D/(D + E)$;
- profitability: $EBITDA/A$;
- size: $\ln(\text{Sales})$.

Interpretation: These controls capture standard capital-structure determinants. The rating-proximity variables are meant to capture an additional, discrete ratings motive.

2.4 POM rating-proximity measure

The POM measure uses plus and minus notches inside broad ratings:

- CRPlus = 1 if the firm has a plus rating, such as $BB+$.
- CRMinus = 1 if the firm has a minus rating, such as $BB-$.
- CRPOM = CRPlus + CRMinus.

Interpretation: A plus rating is closer to a possible upgrade; a minus rating is closer to a possible downgrade. Firms in the middle of a broad rating, such as *BB*, are treated as farther from a broad-rating change.

2.5 Credit Score rating-proximity measure

The paper also builds a Credit Score by regressing numerical ratings on variables that predict ratings, including profitability, leverage, interest coverage, size, and related financial ratios. Within each micro rating, firms are sorted by the score:

- $CR_{High} = 1$ for firms in the top third of their micro rating.
- $CR_{Low} = 1$ for firms in the bottom third of their micro rating.
- $CR_{HOL} = CR_{High} + CR_{Low}$.

Interpretation: This measure captures proximity to a rating change of any kind, not just a broad-rating change. A high-score firm within a rating is closer to upgrade; a low-score firm is closer to downgrade.

3 Models and empirical specifications

3.1 CR-CS intuition

The credit rating–capital structure hypothesis can be summarized as

$$\text{Near rating change} \Rightarrow \text{higher marginal cost of debt} \Rightarrow \text{less net debt relative to equity.} \quad (2)$$

Interpretation: The key point is discreteness. Small debt issuance can be costly for a firm close to a downgrade, even if the same issuance would not change another firm’s rating.

3.2 POM regressions

The baseline POM tests estimate

$$\text{NetDIss}_{it} = \alpha + \beta_0 \text{CRPOM}_{it} + \phi K_{it} + \varepsilon_{it}, \quad (3)$$

$$\text{NetDIss}_{it} = \alpha + \beta_1 \text{CRPlus}_{it} + \beta_2 \text{CRMinus}_{it} + \phi K_{it} + \varepsilon_{it}, \quad (4)$$

$$\text{NetDIss}_{it} = \alpha + \beta_3 \text{CRPOM}_{it} + \varepsilon_{it}. \quad (5)$$

Interpretation: CR-CS predicts negative coefficients. Firms near either an upgrade or a downgrade should issue less net debt relative to equity.

3.3 Credit Score regressions

The Credit Score tests estimate analogous regressions:

$$\text{NetDIss}_{it} = \alpha + \gamma_0 \text{CRHOL}_{it} + \phi K_{it} + \varepsilon_{it}, \quad (6)$$

$$\text{NetDIss}_{it} = \alpha + \gamma_1 \text{CRHigh}_{it} + \gamma_2 \text{CRLow}_{it} + \phi K_{it} + \varepsilon_{it}. \quad (7)$$

Interpretation: The coefficients on high- and low-score firms test whether both upgrade and downgrade concerns reduce debt usage.

3.4 Broad-rating and investment-grade boundary tests

The paper estimates the POM and Credit Score regressions separately by broad rating and separately around the investment-grade/speculative-grade boundary:

$$\text{NetDIss}_{it} = \alpha + \delta \text{CR}_{IG/SG,it} + \phi K_{it} + \varepsilon_{it}. \quad (8)$$

Interpretation: These tests ask whether the effect is concentrated near especially important rating thresholds, such as BBB/BB.

3.5 Nested pecking-order and tradeoff tests

To check whether rating effects are subsumed by traditional theories, the paper nests ratings variables into earlier empirical specifications:

$$\Delta \text{LTD}_{it}/A_{it} = \alpha + \beta \text{DEF}_{it} + \theta \text{CR}_{it} + \varepsilon_{it}, \quad (9)$$

$$\Delta \text{LTD}_{it}/A_{it} = \alpha + \beta (\text{LTD}'_{it} - \text{LTD}_{it}) + \theta \text{CR}_{it} + \varepsilon_{it}. \quad (10)$$

Interpretation: If ratings are merely proxies for pecking-order deficits or tradeoff target adjustments, the rating coefficient should disappear. It does not.

4 Identification and empirical design

4.1 What identifies the effect

- The main comparison is between firms near a rating change and firms in the middle of their rating category.
- The regressions control for standard leverage, profitability, and size variables.
- The tests exclude very large debt offerings because those can affect ratings for nearly all firms.
- The paper also tests rating effects across individual years, broad ratings, and traditional capital-structure frameworks.

Interpretation: The empirical design isolates whether closeness to a discrete rating threshold changes financing behavior, conditional on continuous firm characteristics.

4.2 Why plus/minus ratings help

- Plus and minus notches are observable ex ante.
- They indicate position within a broad rating category.
- They create a simple comparison between edge firms and middle firms inside broad ratings.

Interpretation: The POM test converts the discrete nature of ratings into a clean empirical dummy: edge firms should behave more conservatively.

4.3 Why Credit Scores help

- Not every plus or minus firm is equally close to a future rating change.
- Ratings can change within micro ratings as well as across broad ratings.
- Credit scores use fundamentals to locate firms near the top or bottom of their existing rating.

Interpretation: The Credit Score test is a robustness check that avoids relying only on the agency's plus/minus label.

4.4 Limitations

- Ratings are not randomly assigned.
- Financing decisions may be lumpy and affected by transaction costs.
- Credit ratings may proxy for unobserved firm quality or risk.
- The data measure actual transactions, not the internal financing plans managers consider.
- The sample is restricted to rated firms.

Interpretation: The paper's strength is not random assignment. It is the consistency of the effect across several definitions of rating proximity and several model nests.

5 Tables and figures

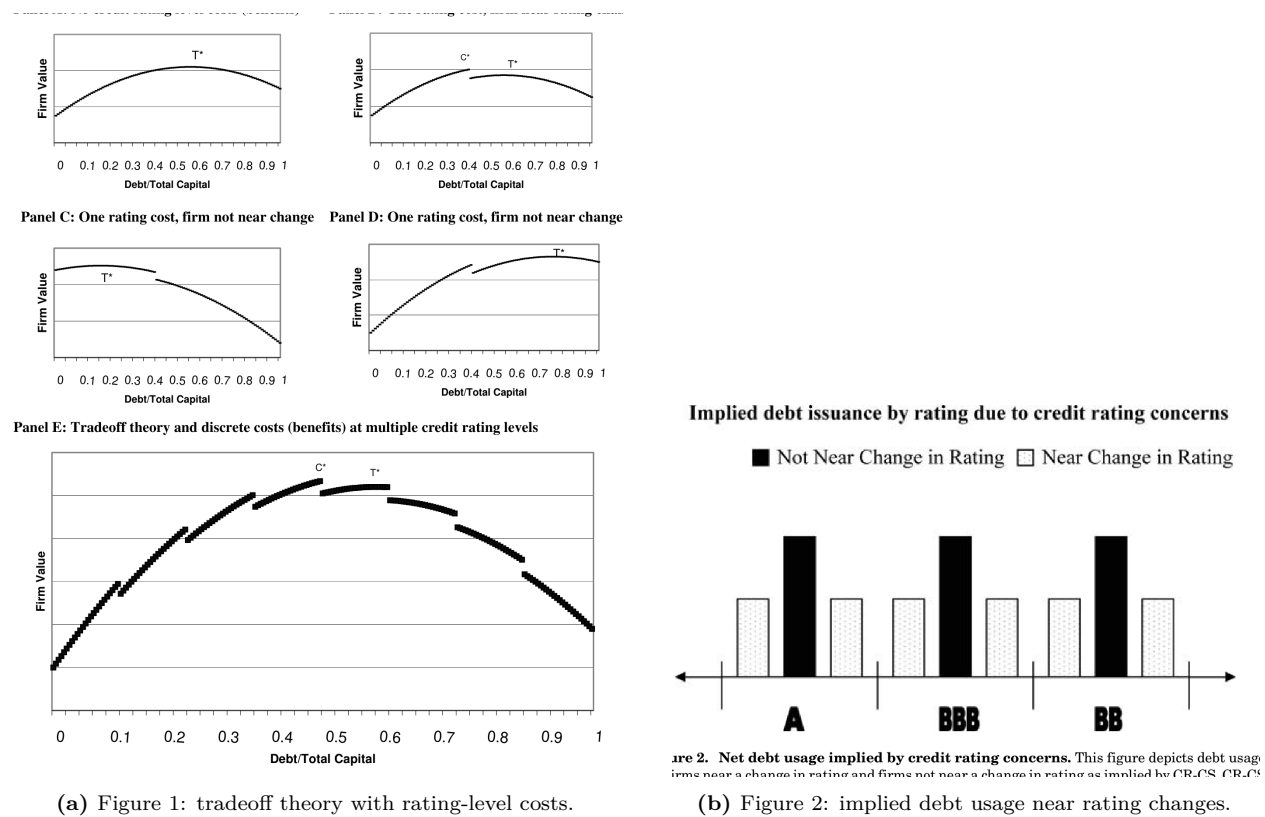
Visuals are directly extracted from the paper and grouped compactly.

5.1 Figure 1 and Figure 2: theory intuition

Read:

- Figure 1 shows how discrete rating costs can create kinks in the firm-value function over leverage.
- Panels B–D illustrate how the optimal leverage ratio can shift away from the tradeoff optimum when a firm is close to a rating change.
- Figure 2 shows the main empirical prediction: firms near a rating change use less debt than firms not near a rating change.

Economic message: Rating thresholds create local incentives. Even if leverage is generally valuable, a firm close to a downgrade may avoid debt because a small movement can trigger a discrete rating cost.



5.2 Table I and Table II: sample and capital activity

Read:

- Table I shows 12,336 firm-years distributed across ratings from AAA to CCC or below.
- Leverage rises as credit quality falls, as expected.
- Table II shows that debt-only transactions are much more common than equity-only transactions.
- Nearly half of firm-years have no capital-market offering or reduction.

Economic message: The sample contains rated firms across the credit-quality spectrum, and capital transactions are lumpy. This motivates excluding very large offerings and focusing on net debt relative to equity.

Sample Summary Statistics—Ratings and Leverage

Means, medians, and standard deviations of debt/(debt + equity) by credit rating within the sample, and the number of firm-years (out of the total sample of 12,336 firms-years) that had the indicated rating at the beginning of the firm-year. The sample is Compustat firms from 1986 to 2001, excluding firms with missing values for regularly used variables in the empirical tests of the paper (these include credit ratings, total assets, debt, and equity). Debt/(debt + equity) is book long-term and short-term debt divided by book long-term and short-term debt plus book shareholders' equity (leverage statistics exclude firms with $D/(D + E)$ greater than 1 or less than 0).

	AAA	AA+	AA	AA−	A+	A
Number of Firm-Years	342	199	622	703	1,135	1,380
Debt/(Debt + Equity)						
Mean	39.3%	30.7%	39.0%	43.2%	46.7%	45.2%
Median	30.5%	37.4%	41.1%	44.1%	46.6%	46.1%
Std Dev.	28.2%	16.7%	18.2%	19.5%	20.5%	18.7%
	A−	BBB+	BBB	BBB−	BB+	BB
Number of Firm-Years	1,067	1,083	1,149	847	521	636
Debt/(Debt + Equity)						
Mean	46.4%	46.9%	48.9%	50.2%	55.4%	55.3%
Median	40.3%	42.1%	45.7%	48.6%	53.1%	53.6%
Std dev.	16.6%	15.9%	16.4%	17.4%	17.9%	17.7%
	BB−	B+	B	B−	CCC+ or Below	
Number of Firm-Years	785	1,067	419	172	209	
Debt/(Debt + Equity)						
Mean	59.5%	63.3%	68.0%	68.0%	62.5%	
Median	57.9%	66.2%	72.1%	70.6%	71.4%	
Std dev.	19.0%	21.0%	18.9%	23.0%	28.6%	

(a) Table I: ratings and leverage.

Sample Summary Statistics—Capital Activity

Number of firm-years in the sample with the indicated capital activity. A debt or equity offering or reduction is defined as the net amount raised or reduced equal to 1% of total assets or greater for the calendar year. The sample is Compustat data covering security issuance from 1986 to 2001 and excludes firms with missing values for regularly used variables in the empirical tests of the paper (these include credit ratings, total assets, debt, and equity).

	Offerings		Reductions	
	N	%	N	%
Debt only	4,829	39.1%	3,264	26.5%
Equity only	896	7.3%	1,657	13.4%
Debt and equity	829	6.7%	641	5.2%
Neither	5,782	46.9%	6,774	54.9%
Total	19,336	100.0%	19,336	100.0%

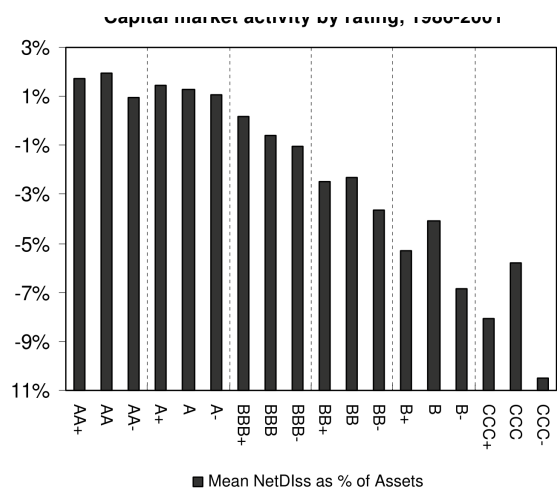
(b) Table II: capital activity.

5.3 Figure 3 and Figure 4: raw financing patterns by rating

Read:

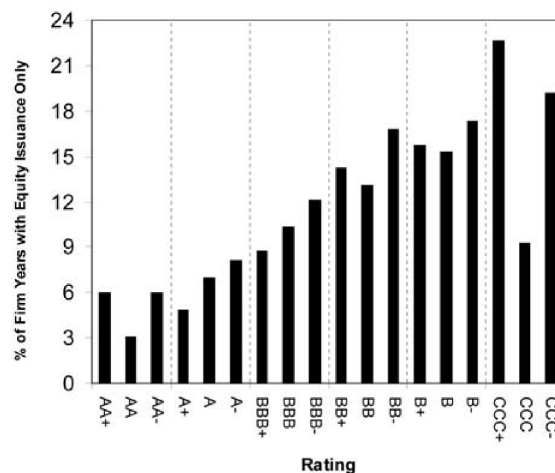
- Figure 3 shows that average net debt issuance minus net equity issuance declines with credit quality deterioration.
- Figure 4 shows that high-quality firms are more likely to issue debt, while lower-rated firms are more likely to issue equity.
- These raw patterns motivate the need for controls in the regression tests.

Economic message: Debt capacity differs sharply by rating. The core tests ask whether edge-of-rating firms behave differently from middle-of-rating firms after accounting for these broad patterns.

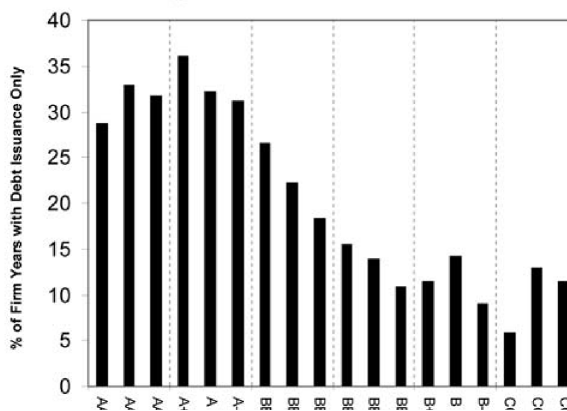


(a) Figure 3: mean NetDIss by rating.

Panel A. Equity offerings, 1986 to 2001



Panel B. Debt offerings, 1986 to 2001



(b) Figure 4: debt and equity offerings by rating.

5.4 Table III: POM tests

Read:

- The coefficient on CRPOM is negative and statistically significant.
- In Panel A, excluding large debt offerings, CRPOM is about -0.0058 with controls and -0.0102 without controls.
- Splitting plus and minus firms shows both CRPlus and CRMinus have negative coefficients.
- Panel B, excluding large debt and equity offerings, produces the same qualitative result.

Economic message: Firms near a broad-rating change issue less debt relative to equity. The effect is economically meaningful and robust to alternative treatment of large offerings.

Credit Rating Impact on Capital Structure Decisions—Plus or Minus Tests

Coefficients and standard errors from pooled time-series cross-section regressions of net debt raised for the year minus net equity raised for the year divided by beginning-of-year total assets on credit rating dummy variables and on control variables measured at the beginning of each year. CR_{POM} is a credit rating dummy variable equal to 1 if the firm has either a plus or a minus credit rating and 0 otherwise. CR_{Plus} and CR_{Minus} are credit rating dummy variables equal to 1 if the firm has a plus or minus rating, respectively and 0 otherwise. The control variables include $D/(D + E)$, book debt divided by book shareholder's equity plus book debt, $EBITDA/A$, previous year's $EBITDA$ divided by total assets, and $\ln(Sales)$, the natural log of total sales. The sample covers security issuance from 1986 to 2001 and excludes observations with missing values for any of the variables. A large offering is defined as an offering greater than 10% of total assets in the year. Errors are White's consistent standard errors. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Panel A: Excluding Large Debt Offerings			Panel B: Excluding Large Debt and Equity Offerings		
	1	2	3	1	2	3
Intercept	-0.0787** (0.0082)	-0.0787** (0.0082)	-0.0006 (0.0012)	-0.0384*** (0.0051)	-0.0384*** (0.0051)	0.0043*** (0.0010)
CR_{POM}	-0.0058*** (0.0016)		-0.0102*** (0.0017)	-0.0027** (0.0013)		-0.0050*** (0.0013)
CR_{Plus}		-0.0064*** (0.0020)			-0.0012 (0.0015)	
CR_{Minus}		-0.0051*** (0.0019)			-0.0044*** (0.0015)	
$D/(D + E)$	-0.0153** (0.0066)	-0.0153** (0.0066)		-0.0095** (0.0045)	-0.0095** (0.0046)	
$EBITDA/A$	0.1288*** (0.0265)	0.1293*** (0.0264)		0.1139*** (0.0123)	0.1133*** (0.0123)	
$\ln(Sales)$	0.0090*** (0.0008)	0.0090*** (0.0008)		0.0042*** (0.0004)	0.0042*** (0.0004)	
Adj. R^2	0.0541	0.0542	0.0030	0.0407	0.0410	0.0015
N	10,842	10,842	10,842	10,573	10,573	10,573

middle.¹⁵ This indicates that the results are not only statistically significant but also economically significant. These results are robust to excluding SIC codes 4000-4999 and 6000-6999 and to examining binary decisions (e.g., debt vs. equity) using logit tests (not reported). The effects are also evident when the dependent variable is broken into net debt only or net equity only.¹⁶

5.5 Table IV: year-by-year POM coefficients

Read:

- The year-by-year regressions show many negative POM coefficients across 1986–2001.
- The strongest negative coefficients appear in several years such as 1987, 1988, 1993, 1994, 1995, and 2001.
- The annual evidence supports the pooled result rather than relying on one unusual year.

Economic message: The rating effect is persistent across the sample period. It is not driven by a single macro episode.

Credit Rating Impact on Capital Structure Decisions—POM Coefficients by Year

Coefficients and standard errors from cross-sectional regressions by year of net debt raised for the year minus net equity raised for the year divided by beginning-of-year total assets on a constant, credit rating dummy variables and control variables measured at the beginning of each year. CR_{POM} is a credit rating dummy variable equal to 1 if the firm has either a plus or a minus credit rating and 0 otherwise. CR_{Plus} and CR_{Minus} are credit rating dummy variables equal to 1 if the firm has a plus or minus rating, respectively and 0 otherwise. Regression 1 includes the CR_{POM} dummy variable and Regression 2 includes the CR_{Plus} and CR_{Minus} credit rating dummy variables. The control variables (not shown) are $D/(D + E)$, book debt divided by book shareholder's equity plus book debt, $EBITDA/A$, $EBITDA$ divided by total assets, and $\ln(Sales)$, the natural log of total sales. The samples exclude observations with missing values for any of the variables. The sample also excludes a firm-year if the firm had a debt offering greater than 10% of total assets in the year. Errors are White's consistent standard errors. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	1986	1987	1988	1989	1990	1991	1992	1993
Regression 1								
CR_{POM}	0.0051 (0.0080)	-0.0173** (0.0084)	-0.0145*** (0.0051)	0.0048 (0.0059)	-0.0043 (0.0047)	-0.0069 (0.0067)	0.0031 (0.0068)	-0.0141** (0.0068)
Regression 2								
CR_{Plus}	0.0002 (0.0092)	-0.0106 (0.0105)	-0.0157*** (0.0063)	0.0016 (0.0068)	0.0022 (0.0057)	-0.0042 (0.0086)	0.0010 (0.0081)	-0.0168** (0.0092)
CR_{Minus}	0.0119 (0.0102)	-0.0249*** (0.0097)	-0.0131** (0.0067)	0.0090 (0.0074)	-0.0110** (0.0065)	-0.0096* (0.0068)	0.0052 (0.0082)	-0.0117* (0.0083)
	1994	1995	1996	1997	1998	1999	2000	2001
Regression 1								
CR_{POM}	-0.0104** (0.0052)	-0.0129** (0.0058)	-0.0052 (0.0067)	-0.0023 (0.0065)	-0.0080* (0.0052)	0.0040 (0.0072)	-0.0046 (0.0060)	-0.0066* (0.0044)
Regression 2								
CR_{Plus}	-0.0132** (0.0063)	-0.0187*** (0.0089)	-0.0062 (0.0086)	-0.0011 (0.0072)	-0.0066 (0.0061)	-0.0029 (0.0092)	-0.0033 (0.0072)	-0.0044 (0.0053)
CR_{Minus}	-0.0077	-0.0064	-0.0039	-0.0037	-0.0096*	0.0122	-0.0060	-0.0091**

5.6 Table V: Credit Score tests

Read:

- The coefficient on CRHOL is negative and significant.
- Splitting the dummy shows both high-score and low-score firms issue less debt relative to equity.
- Using a stricter 5% large-debt cutoff produces very similar results.

Economic message: The result does not depend only on plus/minus ratings. Firms near the top or bottom of a micro rating behave more conservatively as well.

Credit Rating Impact on Capital Structure Decisions—Credit Score Tests

Coefficients and standard errors from pooled time-series cross-section regressions of net debt raised for the year minus net equity raised for the year divided by beginning-of-year total assets on credit rating dummy variables and on various control variables measured at the beginning of each year. CR_{HOL} is a credit rating dummy variable equal to 1 if the firm's Credit Score is in the high or low third of its Micro Rating and 0 otherwise. CR_{High} and CR_{Low} are credit rating dummy variables equal to 1 if the firm's Credit Score is in the high or low third, respectively, within its Micro Rating and 0 otherwise. The control variables include $D/(D + E)$, book debt divided by book shareholder's equity plus book debt, $EBITDA/A$, $EBITDA$ divided by total assets, $\ln(Sales)$, the natural log of total sales. The sample covers security issuance from 1986 to 2001 and excludes observations with missing values for any of the variables. The sample excludes a firm-year if the firm had a debt offering greater than the indicated percentage of total assets in the year. Errors are White's consistent standard errors. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Panel A: Excl. Debt Offerings >10%			Panel B: Excl. Debt Offerings >5%		
	1	2	3	1	2	3
Intercept	-0.0781** (0.0082)	-0.0782** (0.0081)	-0.0009 (0.0013)	-0.0916*** (0.0088)	-0.0894*** (0.0086)	-0.0129*** (0.0013)
CR_{HOL}	-0.0083*** (0.0017)		-0.0091*** (0.0017)	-0.0088*** (0.0017)		-0.0098*** (0.0018)
CR_{High}		-0.0084*** (0.0020)			-0.0075*** (0.0020)	
CR_{Low}		-0.0083*** (0.0025)			-0.0103*** (0.0026)	
$D/(D + E)$	-0.0152** (0.0067)	-0.0153** (0.0068)		-0.0155** (0.0069)	-0.0153** (0.0073)	
$EBITDA/A$	0.1290*** (0.0272)	0.1291*** (0.0277)		0.1047*** (0.0294)	0.1029*** (0.0300)	
$\ln(Sales)$	0.0092*** (0.0008)	0.0092*** (0.0008)		0.0098*** (0.0009)	0.0098*** (0.0009)	
Adj. R^2	0.0554	0.0554	0.0022	0.0589	0.0590	0.0027
N	10,547	10,547	10,547	9,154	9,154	9,154

Source: Compustat, Moody's, and Standard & Poor's. Data for 1986-2001.

5.7 Table VI and Table VII: rating-category thresholds

Read:

- Table VI estimates the rating effect separately by broad rating.
- The effect is negative in several broad-rating groups, with especially strong results in lower ratings for the POM test and in AA/A/BBB/B groups for the Credit Score test.
- Table VII focuses on the investment-grade/speculative-grade boundary.
- Firms near BBB/BB thresholds issue less debt relative to equity even after controlling for the general POM measure.

Economic message: Rating concerns are not confined to a single rating class. The investment-grade boundary is especially natural because many investors and regulations condition on that threshold.

Credit Rating Impact by Broad Rating

Coefficients and standard errors from pooled time-series cross-section regressions by Broad Rating of net debt raised for the year minus net equity raised for the year divided by beginning-of-year total assets on a constant, credit rating dummy variables and control variables measured at the beginning of each year. Regressions 1 and 2 distinguish different credit rating dummy variables included in the regressions. CR_{POM} is a credit rating dummy variable equal to 1 if the firm has either a plus or a minus credit rating and 0 otherwise. CR_{Plus} and CR_{Minus} are credit rating dummy variables equal to 1 if the firm has a plus or minus rating, respectively, and 0 otherwise. CR_{HOL} is a credit rating dummy variable equal to 1 if the firm's Credit Score is in the high or low third of its Micro Rating, and 0 otherwise. CR_{High} and CR_{Low} are credit rating dummy variables equal to 1 if the firm's Credit Score is in the high or low third, respectively, within its Micro Rating and 0 otherwise. The control variables (not shown) are $D/(D + E)$, book debt divided by book shareholder's equity plus book debt, $EBITDA/A$, $EBITDA$ divided by total assets, and $\ln(Sales)$, the natural log of total sales. The sample covers security issuance from 1986 to 2001 and excludes observations with missing values for any of the variables. The sample also excludes a firm-year if the firm had a debt offering greater than 10% of total assets in the year. Errors are White's consistent standard errors. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	AA	A	BBB	BB	B	CCC
Panel A: Plus or Minus Tests						
Regression 1						
CR_{POM}	-0.0052** (0.0024)	0.0004 (0.0022)	0.0020 (0.0027)	-0.0059 (0.0055)	-0.0195*** (0.0077)	-0.0221 (0.0331)
Regression 2						
CR_{Plus}	-0.0028 (0.0034)	0.0007 (0.0025)	0.0064 (0.0031)	-0.0024 (0.0068)	-0.0185*** (0.0078)	-0.0212 (0.0343)
CR_{Minus}	-0.0060** (0.0026)	0.00001 (0.0026)	-0.0037 (0.0033)	-0.0084* (0.0058)	-0.0250** (0.0151)	-0.0263 (0.0510)
Panel B: Credit Score Tests						
Regression 1						
CR_{HOL}	-0.0052** (0.0025)	-0.0062*** (0.0022)	-0.0076*** (0.0028)	-0.0066 (0.0057)	-0.0150** (0.0090)	0.0703 (0.0328)
Regression 2						
CR_{High}	-0.0103*** (0.0032)	-0.0043* (0.0027)	0.0024 (0.0035)	0.0033 (0.0071)	-0.0166* (0.0112)	0.0530 (0.0380)
CR_{Low}	-0.0008 (0.0031)	-0.0079*** (0.0026)	-0.0175*** (0.0035)	-0.0163** (0.0071)	-0.0133 (0.0118)	0.0933 (0.0416)

(a) Table VI: broad-rating tests.

Credit Rating Impact on Capital Structure Decisions—Investment-Grade to Speculative-Grade

Coefficients and standard errors from pooled time-series cross-section regressions of net debt raised for the year minus net equity raised for the year divided by beginning-of-year total assets on credit rating dummy variables and on control variables measured at the beginning of each year. CR_{POM} is a credit rating dummy variable equal to 1 if the firm has either a plus or a minus credit rating and 0 otherwise. $CR_{IG/SG}$ is a credit rating dummy variable equal to 1 if the firm has a rating of BBB- or BBB+ in Panel A or BBB, BBB-, BB+, and BB in Panel B, and 0 otherwise. The control variables include $D/(D + E)$, book debt divided by book shareholder's equity plus book debt, $EBITDA/A$, $EBITDA$ divided by total assets, and $\ln(Sales)$, the natural log of total sales. The sample covers security issuance from 1986 to 2001 and excludes observations with missing values for any of the variables. The sample also excludes a firm-year if the firm had a debt offering greater than 10% of total assets in the year. Errors are White's consistent standard errors. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Panel A: BBB- and BB+		Panel B: BBB, BBB-, BB+, and BB	
	1	2	1	2
Intercept	-0.0826*** (0.0082)	-0.0791** (0.0082)	-0.0814*** (0.0083)	-0.0745 (0.0084)
$CR_{IG/SG}$	-0.0086*** (0.0030)	-0.0068** (0.0031)	-0.0073*** (0.0020)	-0.0091*** (0.0021)
CR_{POM}		-0.0040*** (0.0017)		-0.0071*** (0.0017)
$D/(D + E)$	-0.0153** (0.0066)	-0.0152** (0.0066)	-0.0153*** (0.0066)	-0.0151** (0.0066)
$EBITDA/A$	0.1278*** (0.0265)	0.1278*** (0.0265)	0.1262*** (0.0267)	0.1251*** (0.0268)
$\ln(Sales)$	0.0092*** (0.0008)	0.0090*** (0.0008)	0.0092 (0.0007)	0.0089*** (0.0008)
Adj. R^2	0.0542	0.0546	0.0545	0.0559
N	10,842	10,842	10,842	10,842

DEF is missing;²⁸ firms with SIC codes 4000-4999 and 6000-6999, and I also

(b) Table VII: investment-grade/speculative-grade tests.

5.8 Table VIII and Table IX: nested traditional-theory tests and robustness

Read:

- Table VIII nests rating variables inside pecking-order and tradeoff tests.
- The deficit variable and target-debt adjustment variables retain the expected signs.
- Rating dummies remain negative and significant, meaning CR-CS is not absorbed by these theories.
- Table IX adds many explanatory variables, including intangibles, inventory tax credits, z-score, market-to-book, profitability, SG&A, regulation, depreciation, R&D, tax-loss carryforwards, collateral, and size.
- CRPOM and CRHOL remain negative and significant.

Economic message: Credit ratings add a distinct capital-structure motive. The evidence is not just a relabeling of pecking-order deficits, tradeoff targets, or standard control variables.

Credit Rating Effects in the Context of Various Explanatory Variables

Coefficients and standard errors from pooled time-series cross-section regressions of net debt raised for the year minus net equity raised for the year divided by beginning-of-year total assets on credit rating dummy variables and various explanatory variables. CR_{POM} is a credit rating dummy variable equal to 1 if the firm has either a plus or a minus credit rating. CR_{HOL} is a credit rating dummy variable equal to 1 if the firm's Credit Score is in the high or low third of its Micro Rating. Please see the text for references for each of the additional explanatory variables. All explanatory variables are yearly deltas in the level of the variable, except for the regulation and credit rating dummy variables. Errors are White's consistent standard errors. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	1	2	3
Intercept	-0.0035** (0.0014)	-0.0041*** (0.0015)	-0.0017 (0.0017)
CR_{POM}	-0.0106*** (0.0017)		-0.0102*** (0.0017)
CR_{HOL}		-0.0088*** (0.0017)	-0.0082*** (0.0017)
Intangibles	0.1438*** (0.0483)	0.1462*** (0.0483)	0.1449*** (0.0483)
InvTaxCr	-1.3261** (0.6509)	-1.4003** (0.6551)	-1.3381** (0.6604)
z-Score	-0.0171*** (0.0044)	-0.0172*** (0.0043)	-0.0173*** (0.0044)
MktToBk	-0.0017 (0.0025)	-0.0019 (0.0025)	-0.0020 (0.0025)
Profitability	-0.0895* (0.0493)	-0.0896* (0.0492)	-0.0875* (0.0493)
S,G&A	-0.0001 (0.0001)	-0.0001 (0.0001)	-0.0001 (0.0001)
NonDTxS	0.0113 (0.0417)	0.0134 (0.0417)	0.0113 (0.0416)
RegDum	0.0119*** (0.0016)	0.0105*** (0.0016)	0.0113*** (0.0016)
Depreciation	-0.0715 (0.0745)	-0.0747 (0.0745)	-0.0732 (0.0743)
R&D	0.1453 (0.2443)	0.1378 (0.2432)	0.1430 (0.2436)
TaxLssCar	-0.0696** (0.0267)	-0.0692** (0.0267)	-0.0693** (0.0267)
Collateral	0.0333 (0.0232)	0.0331 (0.0232)	0.0335 (0.0232)
Size	-0.0003 (0.0090)	-0.0005 (0.0090)	-0.0005 (0.0090)
N	10,842	10,547	10,547
R^2	0.0265	0.0254	0.0283

Table VIII
SSM Tests of Pecking Order and Tradeoff Theories with Credit Rating Factors

Coefficients and standard errors from pooled time-series cross-section regressions of net long-term debt raised for the year divided by beginning-of-year total assets on credit rating dummy variables and on DEF or $(LTD - LTD) - LTD$. CR_{HOL} is a credit rating dummy variable equal to 1 if the firm has either a plus or a minus credit rating and 0 otherwise. CR_{HOL} is a credit rating dummy variable equal to 1 if the firm's Credit Score is in the high or low third of its Micro Rating and 0 otherwise. DEF and $(LTD - LTD) - LTD$ are defined as in Shyam-Sunder and Myers (1999): DEF is capital expenditures plus dividend payments plus net increase in working capital and the current portion of long-term debt minus operating cash flow after interest and taxes, and $(LTD - LTD) - LTD$ is target debt level minus current debt level divided by total assets. The sample covers security issuance from 1986 to 2001 and excludes firms with SIC codes 4000-4999 and 6000-6999 and observations with missing values for any of the variables. The sample also excludes a firm-year if the firm had a debt offering greater than 10% of total assets in the year or if the SSM explanatory variable is greater than 10% of total assets for the year. ***, **, and * denote significance at the 1%, 5%, and 10% levels, respectively.

	Panel A: Pecking Order Tests						Panel B: Tradeoff Tests					
	POM Tests			HOL Tests			POM Tests			HOL Tests		
	1	2	3	4	5	6	1	2	3	4	5	6
Intercept	-0.0195*** (0.0006)	-0.0109*** (0.0014)	-0.0160*** (0.0012)	-0.0194*** (0.0008)	-0.0111*** (0.0015)	-0.0166*** (0.0013)	-0.0044*** (0.0007)	-0.0090*** (0.0013)	-0.0013*** (0.0009)	-0.0046*** (0.0009)	-0.0057*** (0.0015)	-0.0016 (0.0017)
DEF	0.5396*** (0.0146)		0.5384*** (0.0146)	0.5453*** (0.0138)		0.5444*** (0.0138)						
$(LTD - LTD)$							0.2255*** (0.0082)		0.2243*** (0.0082)	0.2373*** (0.0086)		0.2357*** (0.0086)
CR_{HOL}		-0.0070*** (0.0018)	-0.0047*** (0.0016)					-0.0069*** (0.0017)	-0.0052*** (0.0015)			
CR_{HOL}				-0.0067*** (0.0020)	-0.0048*** (0.0017)					-0.0077*** (0.0018)	-0.0046*** (0.0018)	
Adj. R^2	0.3130	0.0029	0.3150	0.3165	0.0024	0.3177	0.1154	0.0029	0.1169	0.1198	0.0032	0.2209
N	4,767	4,767	4,767	4,613	4,613	4,613	5,820	5,820	5,820	5,581	5,581	5,581

(a) Table VIII: pecking-order and tradeoff nests.

(b) Table IX: additional explanatory variables.

5.9 Appendix Table AI: why large offerings are special

Read:

- The appendix example shows why firms near a downgrade can issue fewer small debt offerings but more large debt offerings conditional on issuing.
- This explains why the empirical design excludes large debt offerings in the main tests.

Economic message: Large offerings have different rating implications from marginal debt changes, so including them can blur the CR-CS prediction.

Firm Type	General Case			Example, $\gamma = 0.1$		
	Debt, Small	Debt, Large	Do Nothing	Debt, Small (%)	Debt, Large (%)	Do Nothing (%)
A	$\frac{3}{8} - \frac{\gamma}{2} - \frac{\gamma^2}{2}$	$\frac{3}{8} - \frac{\gamma}{2} - \frac{\gamma^2}{2}$	$(\gamma + 0.5)^2$	32.0	32.0	36.0
B	$\frac{3}{8} + \frac{\gamma}{2} - \frac{\gamma^2}{2}$	$\frac{3}{8} - \gamma + \frac{\gamma^2}{2}$	$0.5\gamma + 0.25$	42.0	28.0	30.0

for a Type-B firm the value is $V_B(D, Z)$. Firms of Type A will choose to issue equity rather than debt in more cases than firms of Type B.

6 Short summary

- The paper studies whether credit ratings directly affect capital-structure decisions.
- Credit ratings create discrete costs and benefits because investors, regulations, contracts, and markets respond to rating levels.
- The CR-CS hypothesis predicts less net debt relative to equity for firms near an upgrade or downgrade.
- The paper uses two proximity measures: plus/minus ratings and credit-score thirds within micro ratings.
- The sample covers rated Compustat firms from 1986–2001 and contains 12,336 firm-years.
- Firms near a rating change issue roughly 1% less net debt relative to equity as a share of assets.
- The result appears for plus firms, minus firms, high-score firms, and low-score firms.
- The result is robust across years, broad ratings, the investment-grade boundary, and multiple capital-structure model nests.
- The finding suggests ratings concerns are a distinct managerial consideration in financing policy.

7 Conclusion

7.1 Core conclusion

Kisgen shows that credit ratings directly affect capital-structure decisions. Firms close to rating changes issue less debt relative to equity than firms not close to rating changes, even after controlling for standard determinants of leverage and financing.

7.2 Economic intuition

Ratings matter because the costs and benefits of moving between rating categories are discrete. A small debt issue can trigger a downgrade for one firm but not for another. Managers therefore become more conservative when the firm is near a rating threshold.

7.3 Takeaway

Credit ratings are not merely outcomes of capital structure. They are also inputs into capital-structure decisions. Any empirical or theoretical account of corporate financing for rated firms should incorporate rating thresholds and their discrete effects.